

`raw_materials` app — Cloth Inventory (the factory's input side)

Dual-register guide (developer + owner). Canonical production-subsystem doc: [docs/production/RAW_MATERIALS.md](#) · material price/valuation rules: [ADR-0009 §5](#).

Purpose & business responsibility

Kapde ka stock: kaunsa roll aaya, kahan rakha hai, kis Adda mein use hua, kitna bacha. **Owns:** **cloth identity + consumption facts**. Does NOT own costing math (that's derived later, G1) or production progress.

Models (business + database meaning)

Model	Business	DB / lifecycle	Example row
<code>ClothType</code>	"Cotton", "Rib"	master row, soft <code>is_active</code>	<code>name='Cotton'</code>
<code>ClothColor</code>	"Red - 42 inch"	master row + hex swatch	<code>name='Red'</code> , <code>hex='#c0392b'</code>
<code>StorageLocation</code>	rack/room	master row	<code>name='Rack A'</code>
<code>ClothRoll</code>	ONE physical roll	<code>roll_id='CR-000142'</code> (global sequence, ADR-0010); binds to exactly ONE Adda forever via <code>adda</code> FK; <code>cost_per_kg</code> = PURCHASE price, may be NULL (honest-NULL — never guess)	<code>CR-000142</code> , <code>Cotton/Red</code> , <code>25.40kg</code> , <code>₹200/kg</code> , <code>adda=3-PATTI-001</code> , <code>status=used</code>

Related (lives in production app, written here-adjacent): `LayeringRollEntry` (per-roll verify at attach) and `RemainingClothOfClothRoll` (leftover with weight — MANDATORY at layering completion; reuse recorded ONLY via `consume_leftover`).

How data flows through this app

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intake (BulkRollForm) → ClothRoll rows (status=not_used, price optional)
    | assign_roll_to_adda (LAYERING stage only, weight required)
    ▼
roll.adda set + status=used → layering records weights/layers
    | completion: leftover weigh-in MANDATORY per roll
    ▼
RemainingClothOfClothRoll (provenance: roll + source_adda)
    | physical reuse in another Adda?
    ▼
consume_leftover (sole writer) → is_consumed + consumed_in_adda stamps

```

What records are written & by whom: roll INSERTs + detail edits → `roll_service` (`bulk_create_rolls`, `update_roll_details` — edits are history-logged via tracking) · adda binding → `assign_roll_to_adda` · leftover consumption → `consume_leftover` (C-1; management-only, row-locked).

Why this design / what breaks if bypassed: consumption is the raw fact G1 material-costing will value retroactively. Off-book reuse (leftover used without `consume_leftover`) is UNRECOVERABLE history — Adda B's cloth cost stays understated forever (owner's backfill-known-facts rule forbids inventing it later). Editing `cost_per_kg` as a “market price” silently revalues every historical Adda — it is a purchase FACT (ADR-0009).

Views

Dashboards + roll list (DataTables, stacked cards on phone — F1 fixed), bulk intake form (price fields visible to financial roles only), roll detail.

Common mistakes

1. Don't set `is_consumed` / `consumed_in_adda` by hand — only the service.
2. Don't “update” `cost_per_kg` to today's market rate — corrections only.
3. Unpriced roll ≠ free roll: NULL surfaces on the costing dashboard banner.
4. A roll never splits across Addas — leftovers are the split mechanism.

Django Learning Notes

- **Sequence-generated IDs** (`CR-XXXXXX`): `editable=False`; globally unique forever (ADR-0010 §1 — factory #2 shares the sequence).

- **PROTECT on roll FKs:** a roll referenced by layering entries can't be deleted — physical history.
- **Role-gated form fields:** the intake form POPS `cost_per_kg` for non-financial users (`fields.pop`) — gating belongs in the form layer, display mein nahi.
- **Honest-NULL:** nullable Decimal + dashboard surfacing instead of default-0 — “pata nahi” aur “zero” alag cheez hai.

Related ADRs: 0009 (price semantics, leftover write path) · 0010 (global IDs).

Real factory example

20 rolls aaye Cotton Red ke. Intake → CR-000201..220 (accountant ne ₹190/kg bhara). Layering pe 3 roll 3-PATTI-002 se bind hue (weight verify), complete pe har roll ka bacha hua tukda weigh hua (RemainingCloth rows). Mahine baad ek leftover 3-PATTI-005 mein lag gaya → manager ne consume_leftover se record kiya. G1 aane par: 3-PATTI-002 ka cloth cost = (attach weight – leftovers) × ₹190 — bina kisi backfill ke, kyunki facts pehle se sahi likhe the.